

L2 - 13.5. Basic Dynamic Analysis

Task 1 - Introduction

Malware can use techniques to hide its features from a malware analyst. But no matter how good malware hides its features from static analysis, its primary purpose is to execute. And when malware executes, it leaves traces that a malware analyst can use to identify if it's malicious.

Task 2 - Sandboxing

Static analysis examines malware without executing it, while dynamic analysis observes its behavior during real-time execution. Dynamic analysis requires a strictly sandboxed, controlled lab environment because it intentionally executes malware, posing a far higher risk than static analysis.

The following setup will be required to create a sandbox:

- An isolated machine, ideally a lab machine, that is not connected to live or production systems and is dedicated to malware analysis.
- The ability of the isolated or lab machine to save its initial clean state and revert to that state once malware analysis is complete. This functionality is often called creating and reverting a snapshot. We will need to revert to the original clean state before analyzing a new malware so that infection from the previous malware doesn't contaminate the analysis of the next one.
- Monitoring tools that help us analyze the malware while it's executing inside the Lab Machine. These tools can be automated, as we see in automated sandboxes, or they can be manual, requiring the analyst to interact while performing analysis. We will learn about some of these tools later in the room.
- A file-sharing mechanism that can be used to introduce the malware into the Lab Machine and sends the analysis data or reports out to us. Often, shared directories or network drives are used for this purpose. However, we must be careful that the shared directory is unmounted when executing the malware,

as the malware might infect all the files. This is especially true of ransomware, which might encrypt all shared drives or directories.

Virtualization: A lot of commercial and free tools are available for virtualization. Some of the most famous ones include Oracle's VirtualBox and VMware's Player and Workstation. These three tools allow us to create Lab Machines isolated from our local machine. However, VMWare Player can't create snapshots. For dynamic analysis of malware, snapshot creation is a critical requirement, which makes VMWare Player unsuitable for malware analysis. VMWare Workstation and VirtualBox have the snapshot creation option and are, therefore, suitable for malware analysis. VirtualBox is free, but VMWare Workstation has a paid license.

Apart from these, server-based virtualization software like XenServer, QEmu, ESXi, etc., help with virtualization on a dedicated server. This type of setup is often used by enterprises for their virtualization needs. Security research organizations often use similar technologies to create a VM farm for large-scale virtualization.

Analysis Tools: Automated malware analysis systems have some built-in tools that analyze malware behaviour. For example, in Cuckoo's sandbox, cuckoomon is a tool that records malware activity in a Cuckoo sandbox setup. Once we have our required tools installed on the VM and before running any malware on the VM, we must take a snapshot. After analysis of every malware, we must revert the VM to this snapshot, which will hold the clean state of the VM. This will ensure that our analysis is not contaminated by different malware samples running simultaneously.

File-sharing: Different platforms provide different options for sharing files between host and guest OS. In the most popular tools, i.e., Oracle VirtualBox or VMWare Workstation, the following options are common:

- Shared folder.
- Creating an iso in the host and mounting it to the VM.
- Clipboard copy and paste.

Once we have created a VM, set up analysis tools, taken a snapshot, and placed the malware inside our sandbox, we can start analysing our malware.

Task 3 - ProcMon

ProcMon is part of the Sysinternals suite, a set of utilities created by a company named Winternals Software and purchased by Microsoft in 2006. Sysinternals consists of many handy utilities that provide advanced functionalities for Windows. Sysinternals utilities are widely used in Security research.

Process Monitor - Sysinternals: www.sysinternals.com

File Edit Event Filter Tools Options Help

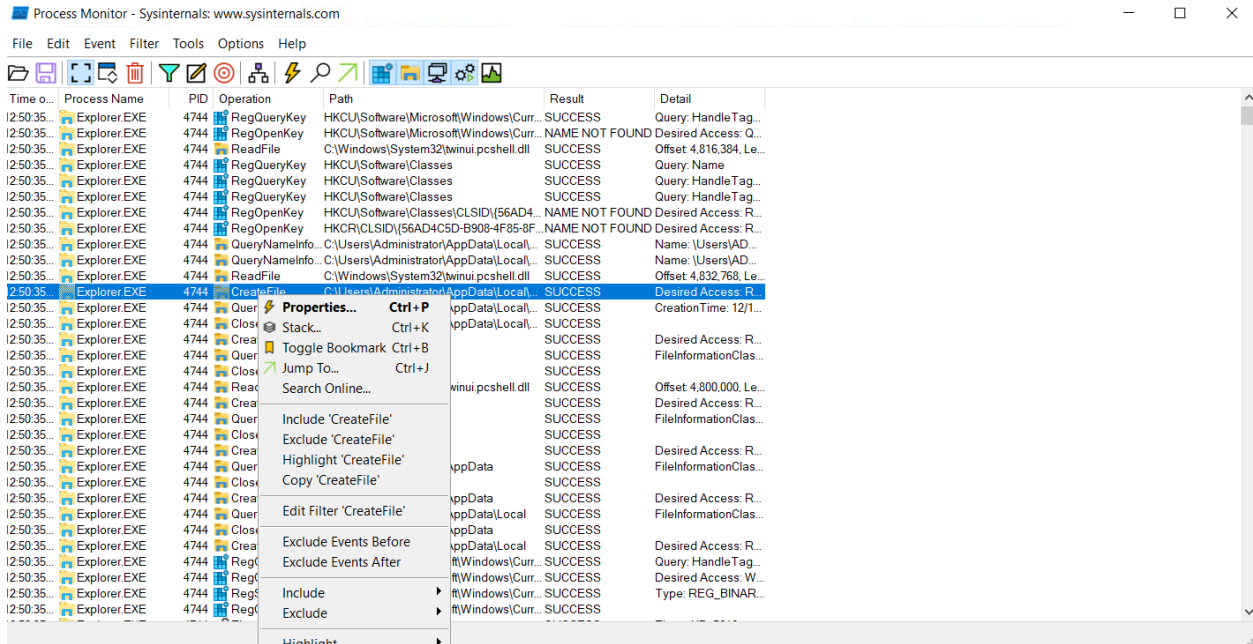
Time ... Process Name PID Operation Path Result Detail

Time ...	Process Name	PID	Operation	Path	Result	Detail
12:50:35...	Explorer.EXE	4744	RegQueryValue	HKCU\Software\Microsoft\Windows\CurrentVersion\...	SUCCESS	Query: HandleTag...
12:50:35...	Explorer.EXE	4744	RegOpenKey	HKCU\Software\Microsoft\Windows\CurrentVersion\...	NAME NOT FOUND	Desired Access: R...
12:50:35...	Explorer.EXE	4744	ReadFile	C:\Windows\System32\winui.pcsHELL.dll	SUCCESS	Offset: 4,816,384, Le...
12:50:35...	Explorer.EXE	4744	RegQueryValue	HKCU\Software\Classes	SUCCESS	Query: Name
12:50:35...	Explorer.EXE	4744	RegQueryValue	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
12:50:35...	Explorer.EXE	4744	RegQueryValue	HKCU\Software\Classes	SUCCESS	Query: HandleTag...
12:50:35...	Explorer.EXE	4744	RegOpenKey	HKCU\Software\Classes\CLSID\{56AD4...}	NAME NOT FOUND	Desired Access: R...
12:50:35...	Explorer.EXE	4744	RegOpenKey	HKCR\CLSID\{56AD4C5D-B908-4F85-BF...	NAME NOT FOUND	Desired Access: R...
12:50:35...	Explorer.EXE	4744	QueryNameInfo	C:\Users\Administrator\AppData\Local\...	SUCCESS	Name: \Users\AD...
12:50:35...	Explorer.EXE	4744	QueryNameInfo	C:\Users\Administrator\AppData\Local\...	SUCCESS	Name: \Users\AD...
12:50:35...	Explorer.EXE	4744	ReadFile	C:\Windows\System32\winui.pcsHELL.dll	SUCCESS	Offset: 4,832,768, Le...
12:50:35...	Exp...		Properties...	C:\Users\Administrator\AppData\Local\...	SUCCESS	Desired Access: R...
12:50:35...	Exp...		Stack...	C:\Users\Administrator\AppData\Local\...	SUCCESS	CreationTime: 12/1...
12:50:35...	Exp...		Toggle Bookmark	C:\Users\Administrator\AppData\Local\...	SUCCESS	
12:50:35...	Exp...		Jump To...	C:\	SUCCESS	Desired Access: R...
12:50:35...	Exp...		Search Online...	C:\Users	SUCCESS	FileInformationClas...
12:50:35...	Exp...			C:\	SUCCESS	
12:50:35...	Exp...			C:\Windows\System32\winui.pcsHELL.dll	SUCCESS	Offset: 4,800,000. Le...
12:50:35...	Exp...			C:\Users	SUCCESS	Desired Access: R...
12:50:35...	Exp...			C:\Users\ADMIN~1	SUCCESS	FileInformationClas...
12:50:35...	Exp...			C:\Users	SUCCESS	
12:50:35...	Exp...			C:\Users\Administrator	SUCCESS	Desired Access: R...
12:50:35...	Exp...			C:\Users\Administrator\AppData	SUCCESS	FileInformationClas...
12:50:35...	Exp...			C:\Users\Administrator	SUCCESS	
12:50:35...	Exp...			C:\Users\Administrator\AppData	SUCCESS	Desired Access: R...
12:50:35...	Exp...			C:\Users\Administrator\AppData\Local	SUCCESS	FileInformationClas...
12:50:35...	Exp...			C:\Users\Administrator\AppData	SUCCESS	
12:50:35...	Exp...			C:\Users\Administrator\AppData\Local	SUCCESS	Desired Access: R...
12:50:35...	Exp...			HKCU\Software\Microsoft\Windows\CurrentVersion\...	SUCCESS	Query: HandleTag...
12:50:35...	Exp...			HKCU\Software\Microsoft\Windows\CurrentVersion\...	SUCCESS	Desired Access: W...
12:50:35...	Exp...			HKCU\Software\Microsoft\Windows\CurrentVersion\...	SUCCESS	Type: REG_BINAR...
12:50:35...	Exp...			HKCU\Software\Microsoft\Windows\CurrentVersion\...	SUCCESS	

Context Menu:

- Properties... Ctrl+P
- Stack... Ctrl+K
- Toggle Bookmark Ctrl+B
- Jump To... Ctrl+J
- Search Online...
- Include 'Explorer.EXE'
- Exclude 'Explorer.EXE'
- Highlight 'Explorer.EXE'
- Copy 'Explorer.EXE'
- Edit Filter 'Explorer.EXE'
- Exclude Events Before
- Exclude Events After
- Include
- Exclude
- Highlight

options are related to filtering. For example, if we choose the option **Include 'Explorer.EXE'**, ProcMon will only show events with Process Name Explorer.E
we choose the option **Exclude 'Explorer.EXE'**, it will exclude Explorer.EXE from



Process Tree: ProcMon also allows us to view all the existing processes in a parent-child relationship, forming a process tree. This can be done by clicking icon in the menu. This option helps identify the parents and children of different processes. As shown by ProcMon, an example process tree can be seen below.

Process Tree

☐ Only show processes still running at end of current trace

☒ Timelines cover displayed events only

Process	Description	Image Path	Life Time	Company	Owner
svchost.exe (1752)	Host Process for ...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
svchost.exe (860)	Host Process for ...	C:\Windows\Syste...		Microsoft Corporati...	NT AUTHOR
TrustedInstaller.exe (1836)	Windows Modules...	C:\Windows\servic...		Microsoft Corporati...	NT AUTHOR
lsass.exe (796)	Local Security Aut...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
fontdrvhost.exe (944)	Usermode Font Dr...	C:\Windows\sys...		Microsoft Corporati...	Font Driver H
csrss.exe (652)	Client Server Runti...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
winlogon.exe (708)	Windows Logon A...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
fontdrvhost.exe (936)	Usermode Font Dr...	C:\Windows\sys...		Microsoft Corporati...	Font Driver H
LogonUI.exe (808)	Windows Logon U...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
dwm.exe (912)	Desktop Window ...	C:\Windows\sys...		Microsoft Corporati...	Window Man:
csrss.exe (4452)	Client Server Runti...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
winlogon.exe (4496)	Windows Logon A...	C:\Windows\sys...		Microsoft Corporati...	NT AUTHOR
fontdrvhost.exe (4564)	Usermode Font Dr...	C:\Windows\sys...		Microsoft Corporati...	Font Driver H
dwm.exe (4616)	Desktop Window ...	C:\Windows\sys...		Microsoft Corporati...	Window Man:
Explorer.EXE (4744)	Windows Explorer	C:\Windows\Explo...		Microsoft Corporati...	WIN-7BS0GF
Procmon.exe (1692)	Process Monitor	C:\ProgramData\c...		Sysinternals - www...	WIN-7BS0GF
Procmon64.exe (1740)	Process Monitor	C:\Users\ADMINI~...		Sysinternals - www...	WIN-7BS0GF
GoogleCrashHandler.exe (5960)	Google Crash Han...	C:\Program Files (...)		Google LLC	NT AUTHOR
GoogleCrashHandler64.exe (5968)	Google Crash Han...	C:\Program Files (...)		Google LLC	NT AUTHOR

Description:

Company:

Path: Idle

Command:

User:

PID: 0 Started: 12/19/2022 12:38:49 AM

Go To Event Include Process Include Subtree Close

Answer the questions below

Monitor the sample ~Desktop\Samples\1.exe using ProcMon. This sample makes a few network connections. What is the first URL on which a network connection is made?

94-73-155-12.cizgi.net.tr:2448 (right click **Include 'Explorer.EXE'**)

What network operation is performed on the above-mentioned URL?

TCP Reconnect

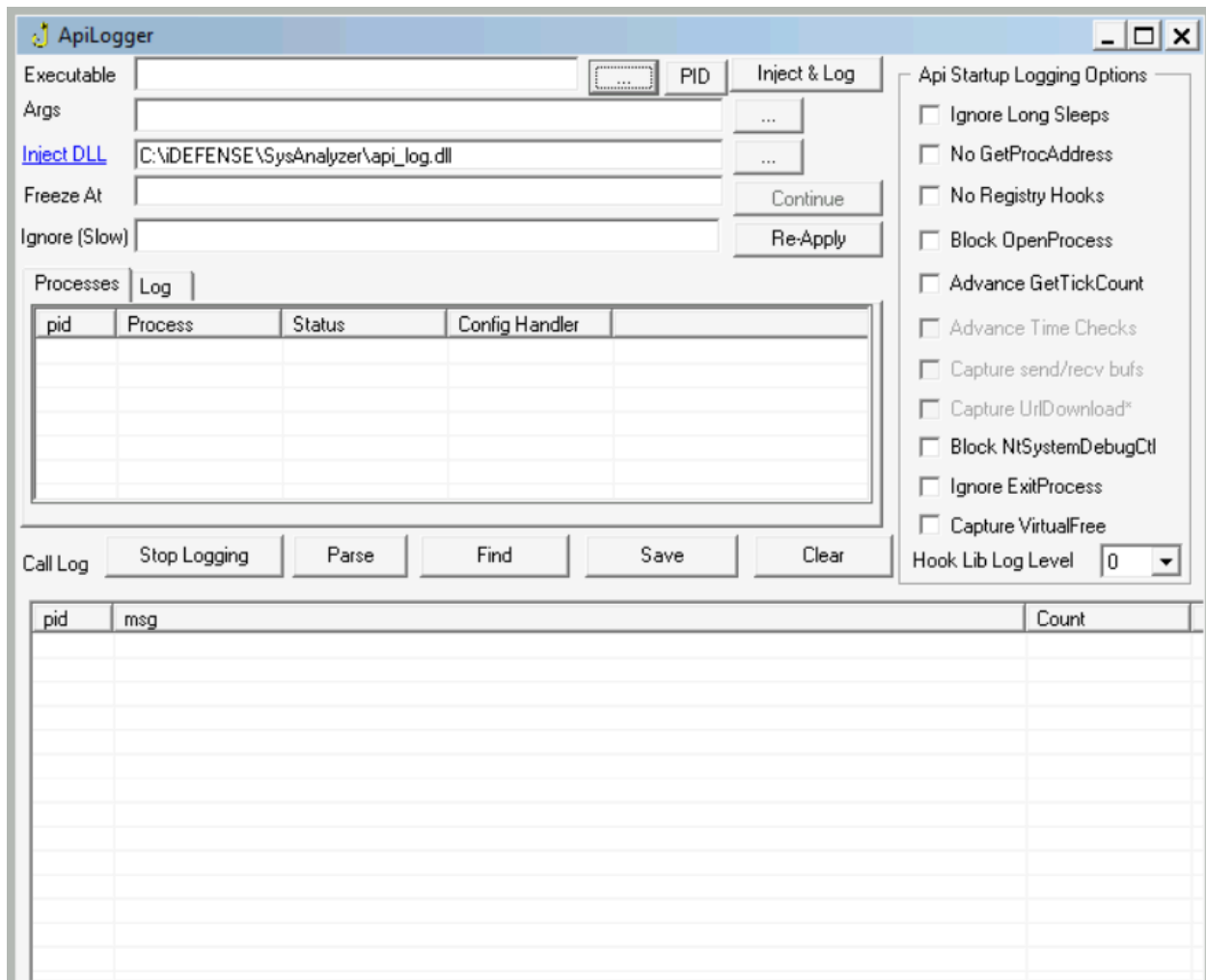
What is the name with the complete full path of the first process created by this sample?

C:\Users\Administrator\Desktop\samples\1.exe

Task 4 - API logger and API monitor

The Windows OS abstracts the hardware and provides an Application Programmable Interface (API) for performing all tasks. For example, there is an API for creating files, an API for creating processes, an API for creating and deleting registries and so on. Therefore, one way to identify malware behaviour is to monitor which APIs a malware calls. The names of the APIs are generally self-explanatory.

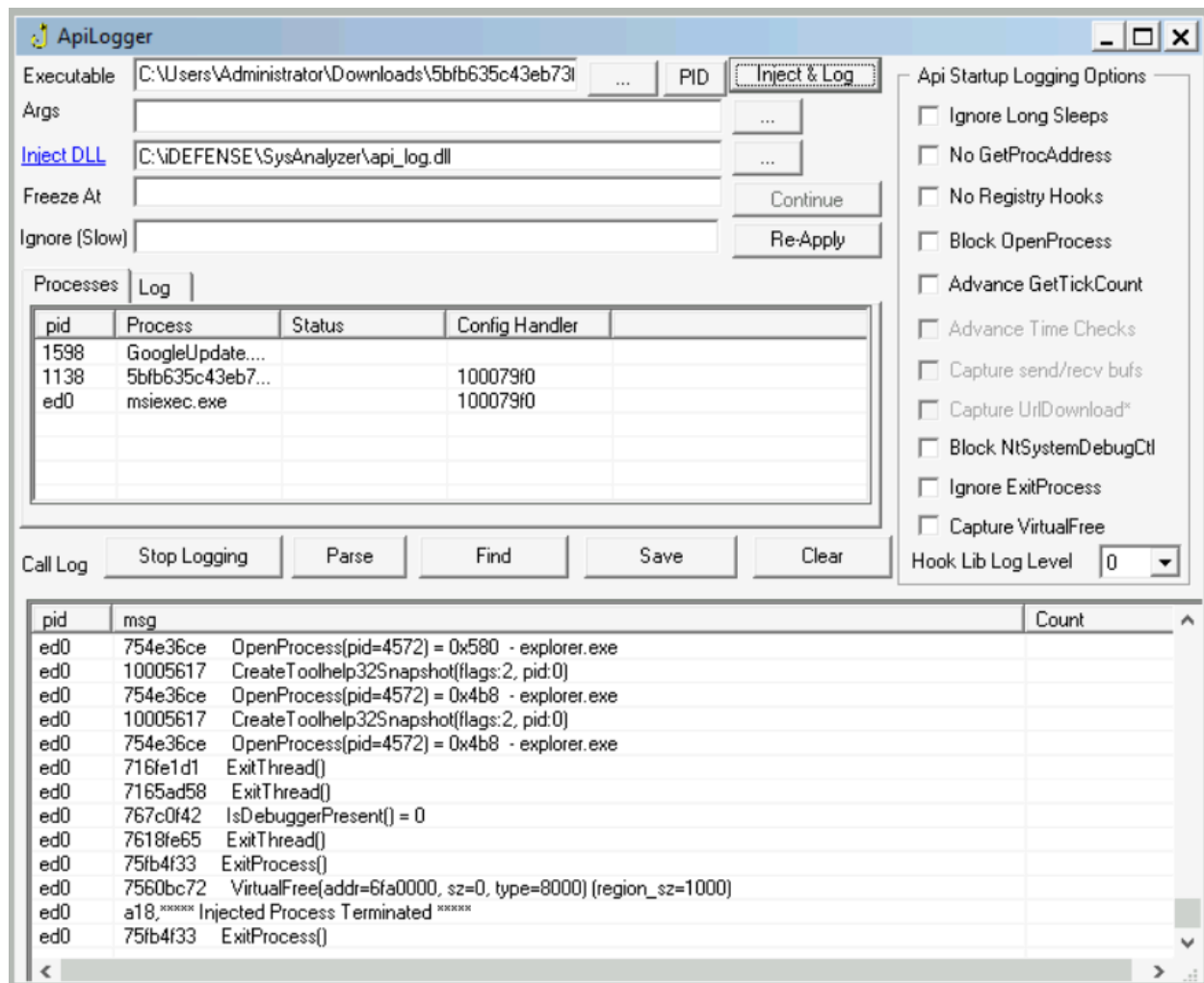
API Logger: The API Logger is a simple tool that provides basic information about APIs called by a process



To open a new process, we can click the highlighted three-dot menu. When clicked, a file browser allows us to select the executable for which we want to monitor the API calls. Once we select the executable, we can click 'Inject & Log' to start the API logging process. We will see the log of API calls in the lower pane, as seen in the picture below. In the upper pane, we see the running processes and their PIDs.

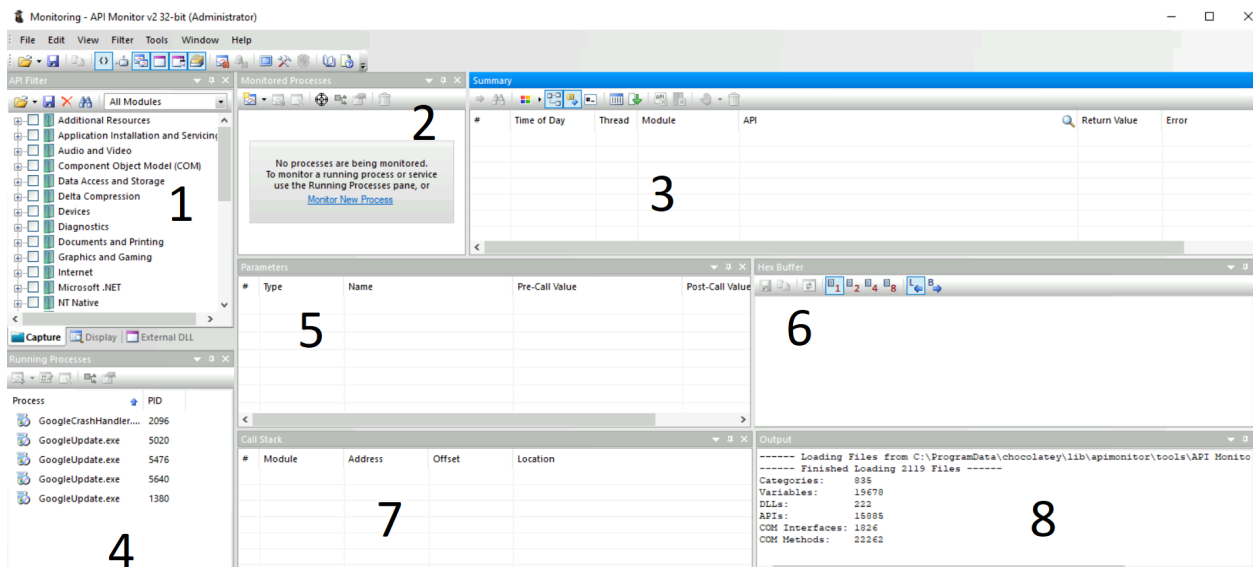
We can see the PID of the process we monitor and the API called with basic information about the API in the 'msg' field.

We can click the 'PID' menu for the API logger to log API calls of a running process. It will open the following window.

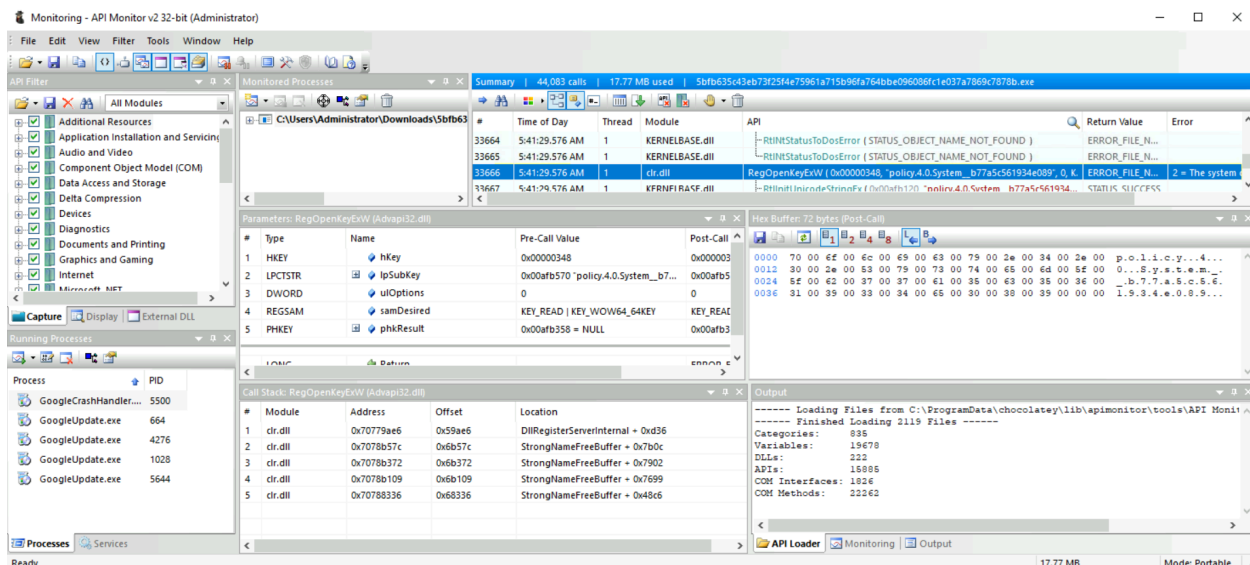


API Monitor:

The API Monitor provides more advanced information about a process's API calls. API Monitor has 32-bit and 64-bit versions for 32-bit and 64-bit processes, respectively.



1. This tab is a filter for the API group we want to monitor. For example, we have a group for 'Graphics and Gaming' related APIs, another for 'Internet' related APIs and so on. API Monitor will only show us APIs from the group we select from this menu.
2. This tab shows the processes being monitored for API calls. We can click the 'Monitor New Process' option to start monitoring a new process.
3. This tab shows the API call, the Module, the Thread, Time, Return Value, and any errors. We can monitor this tab for APIs called by a process.
4. This tab shows running processes that API Monitor can monitor.
5. This tab shows the Parameters of the API call, including the values of those Parameters before and after the API calls.
6. This tab shows the Hex buffer of the selected value.
7. This tab shows the Call Stack of the process.
8. Finally, this tab shows the Output.



- In Tab 1, we see that we have selected all values so that we can monitor all the API calls.
- In Tab 2, we see the path of the process we are monitoring.
- In Tab 3, we see a summary of the API calls. The highlighted API call can be seen as RegOpenKeyExW. Hence we know that the process tried to open a registry key. We see that the API call returns an error, which we can see in the 'Return Value' field of this tab, and the error details can be found in this tab's 'Error' field.
- Tab 5 shows the parameters of the API call from before and after the API call was made.
- Tab 6 shows the selected value in Hex.
- Tab 7 shows the Call Stack of the process.

Answer the questions below

The sample ~Desktop\samples\1.exe creates a file in the C:\ directory. What is the name with the full path of this file?

C:\myapp.exe

What API is used to create this file?

CreateFileA

In Question 1 of the previous task, we identified a URL to which a network connection was made. What API call was used to make this connection?

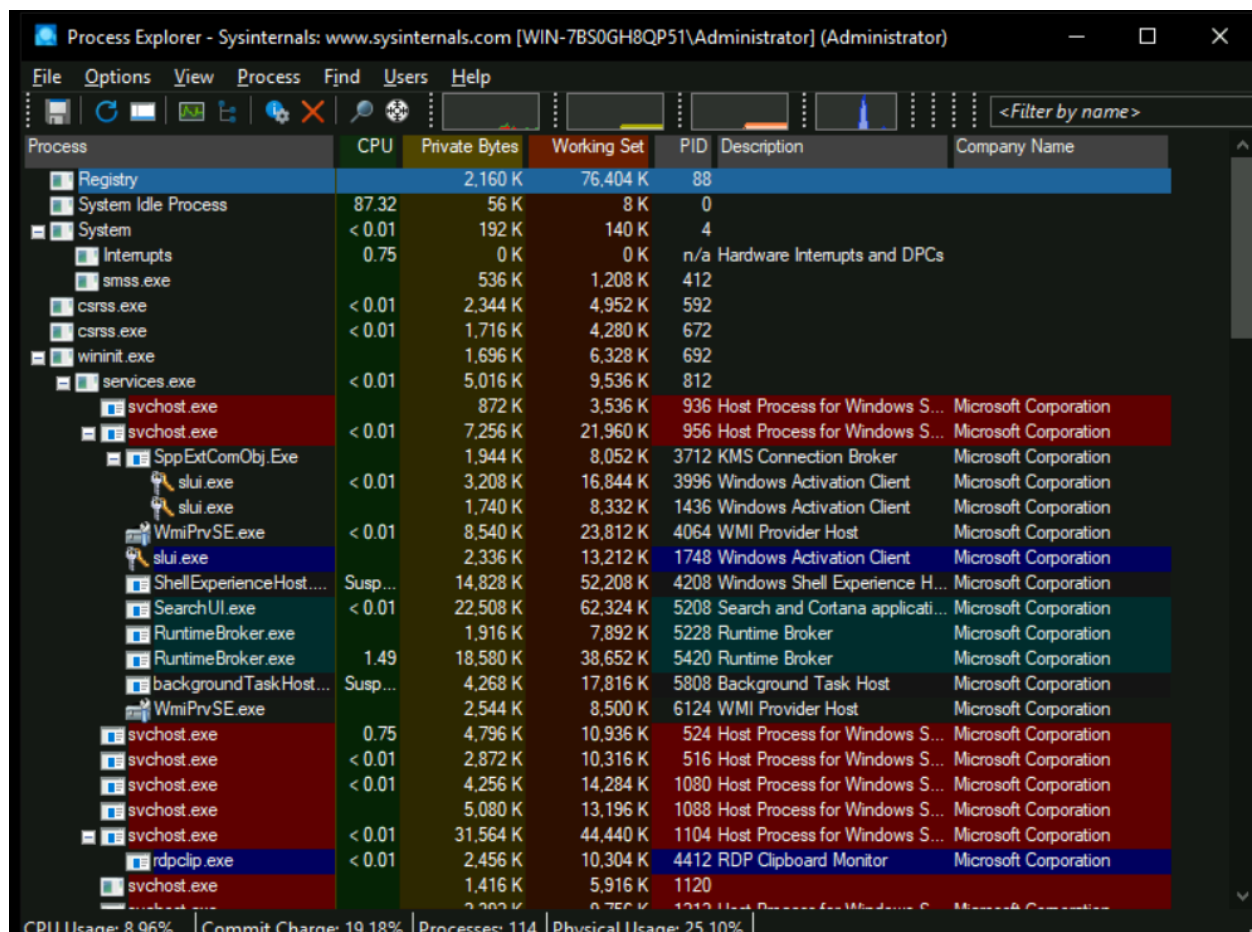
InternetConnectW (Click find , write ip from task 1)

We noticed in the previous task that after some time, the sample's activity slowed down such that there was not much being reported against the sample. Can you look at the API calls and see what API call might be responsible for it?

Sleep

Task 5 - Process Explorer

Process Explorer is another very useful tool from the Sysinternals Suite. It can be considered a more advanced form of the Windows Task Manager. Process Explorer is a very powerful tool that can help us identify process hollowing and masquerading techniques.

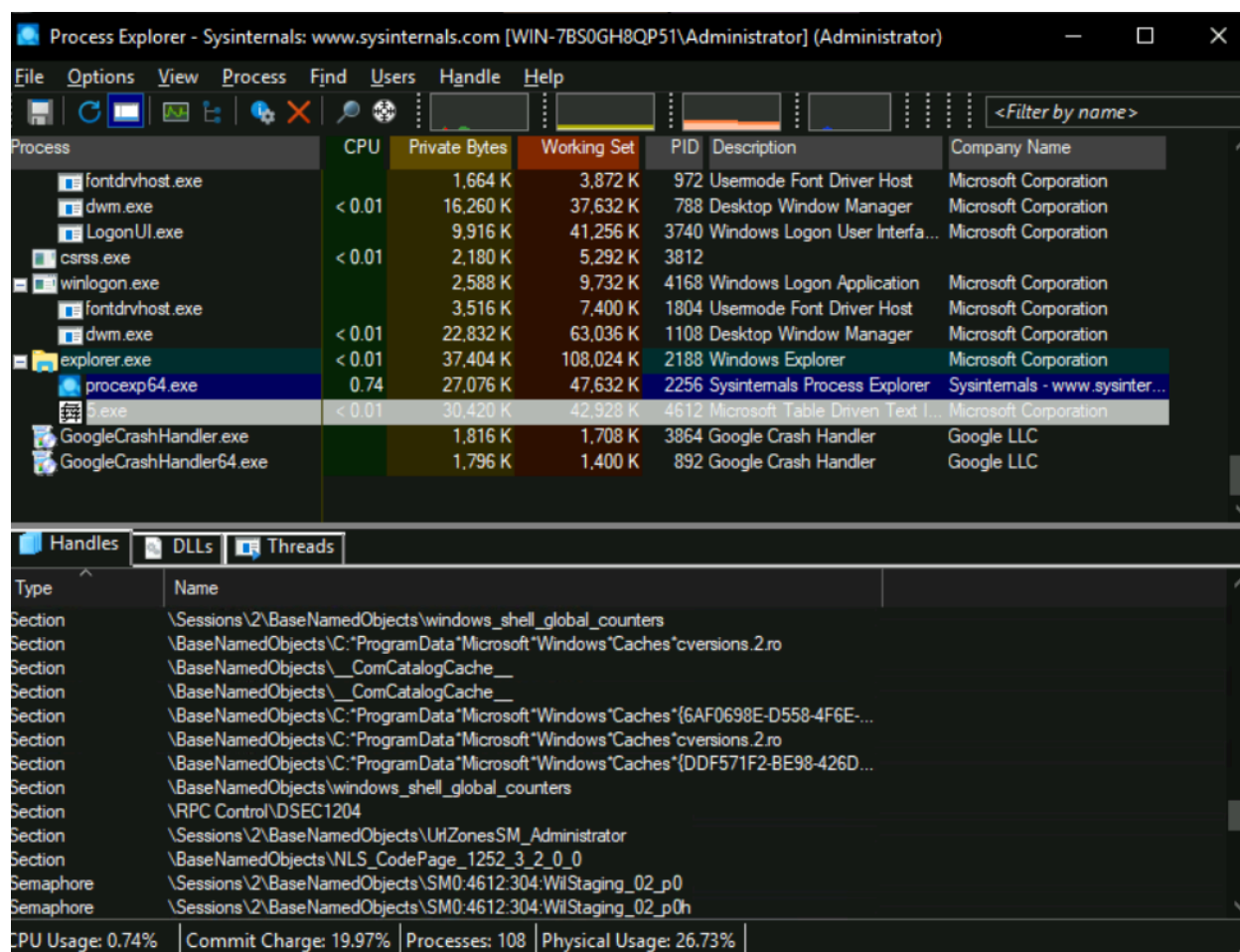


Process Explorer - Sysinternals: www.sysinternals.com [WIN-7BS0GH8QP51\Administrator] (Administrator)

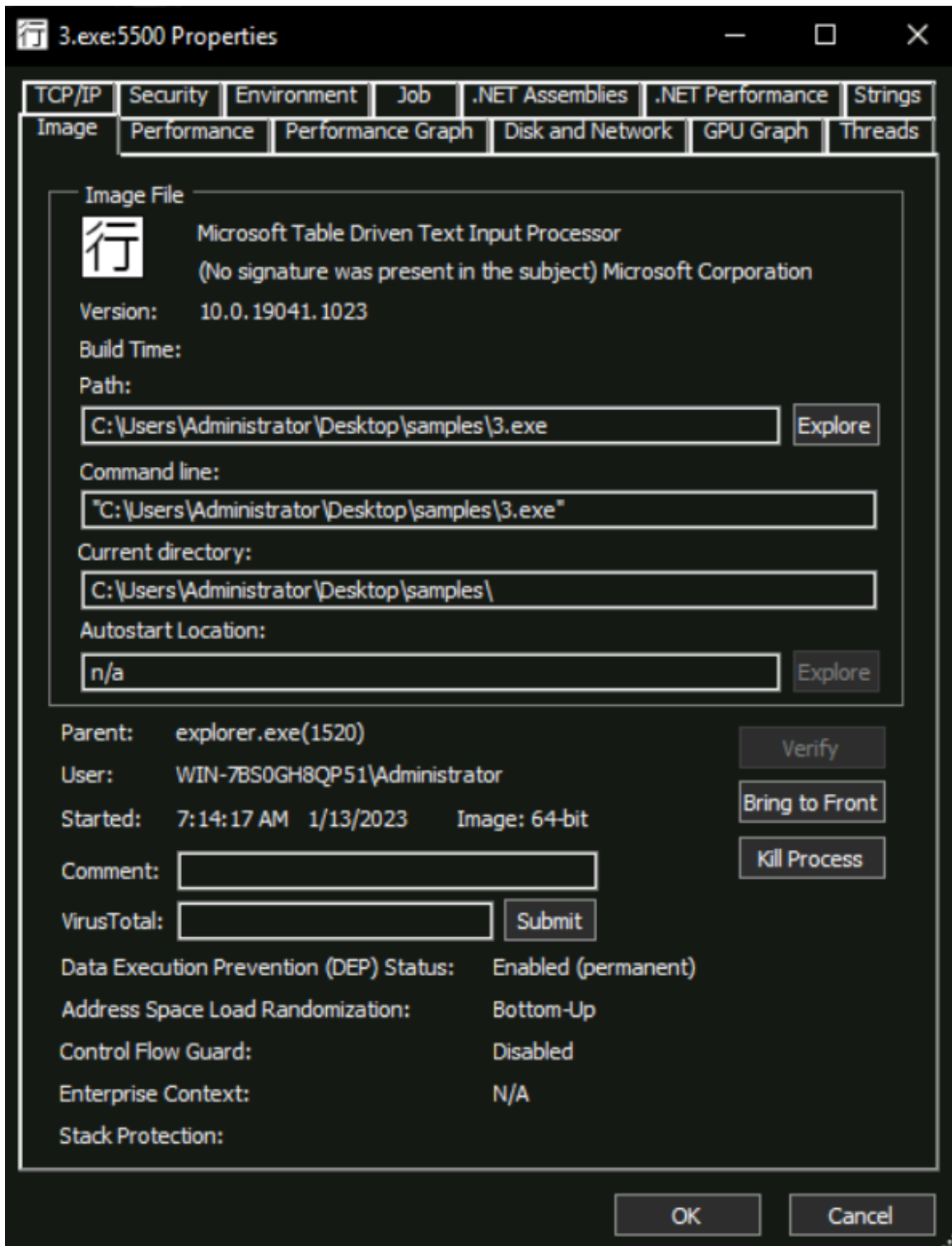
Process	CPU	Private Bytes	Working Set	PID	Description	Company Name
Registry		2,160 K	76,404 K	88		
System Idle Process	87.32	56 K	8 K	0		
System	< 0.01	192 K	140 K	4		
Interrupts	0.75	0 K	0 K	n/a	Hardware Interrupts and DPCs	
smss.exe		536 K	1,208 K	412		
csrss.exe	< 0.01	2,344 K	4,952 K	592		
csrss.exe	< 0.01	1,716 K	4,280 K	672		
wininit.exe		1,696 K	6,328 K	692		
services.exe	< 0.01	5,016 K	9,536 K	812		
svchost.exe		872 K	3,536 K	936	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	7,256 K	21,960 K	956	Host Process for Windows S...	Microsoft Corporation
SppExtComObj.Exe		1,944 K	8,052 K	3712	KMS Connection Broker	Microsoft Corporation
slui.exe	< 0.01	3,208 K	16,844 K	3996	Windows Activation Client	Microsoft Corporation
slui.exe		1,740 K	8,332 K	1436	Windows Activation Client	Microsoft Corporation
WmiPrvSE.exe	< 0.01	8,540 K	23,812 K	4064	WMI Provider Host	Microsoft Corporation
slui.exe		2,336 K	13,212 K	1748	Windows Activation Client	Microsoft Corporation
ShellExperienceHost....	Susp...	14,828 K	52,208 K	4208	Windows Shell Experience H...	Microsoft Corporation
SearchUI.exe	< 0.01	22,508 K	62,324 K	5208	Search and Cortana applicati...	Microsoft Corporation
RuntimeBroker.exe		1,916 K	7,892 K	5228	Runtime Broker	Microsoft Corporation
RuntimeBroker.exe	1.49	18,580 K	38,652 K	5420	Runtime Broker	Microsoft Corporation
backgroundTaskHost....	Susp...	4,268 K	17,816 K	5808	Background Task Host	Microsoft Corporation
WmiPrvSE.exe		2,544 K	8,500 K	6124	WMI Provider Host	Microsoft Corporation
svchost.exe	0.75	4,796 K	10,936 K	524	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	2,872 K	10,316 K	516	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	4,256 K	14,284 K	1080	Host Process for Windows S...	Microsoft Corporation
svchost.exe		5,080 K	13,196 K	1088	Host Process for Windows S...	Microsoft Corporation
svchost.exe	< 0.01	31,564 K	44,440 K	1104	Host Process for Windows S...	Microsoft Corporation
rdpclip.exe	< 0.01	2,456 K	10,304 K	4412	RDP Clipboard Monitor	Microsoft Corporation
svchost.exe		1,416 K	5,916 K	1120		

CPU Usage: 8.96% | Commit Charge: 19.18% | Processes: 114 | Physical Usage: 25.10%

The above screenshot shows all the different processes running in the system in a tree format. We can also see their CPU utilization, memory usage, Process IDs (PIDs), Description, and Company name. We can enable the lower pane view from the 'View' menu to find more information about the processes. When enabled, we see the following screenshot.



When we select a process in the upper pane, we can see details about that process in the lower pane. Here, we see the Handles the process has opened for different Sections, Processes, Threads, Files, Mutexes, and Semaphores. Handles inform us about the resources being used in this process. If another process or a thread in another process is opened by a process, it can indicate code injection into that process. Similarly, we can see DLLs and Threads of the process in the other tabs of the lower pane.



Process Masquerading: Detecting fake binaries: Digital signature verification exposes unsigned malicious executables pretending to be legitimate software like

Microsoft. Limitation with process hollowing: Disk image verification fails to catch malicious code injected into memory, requiring further memory-level analysis.

Process Hollowing: Memory-based code replacement: Malware replaces a legitimate running process's memory code with malicious code while preserving the process identity. Bypassing basic checks: Process Explorer's Strings tab helps detect hollowing where standard disk-level signature verification fails.

Answer the questions below

What is the name of the first Mutex created by the sample
~Desktop\samples\1.exe? If there are numbers in the name of the Mutex,
replace them with X.

\Sessions\X\BaseNamedObjects\SMX:XXXX:XXX:WilStaging_XX

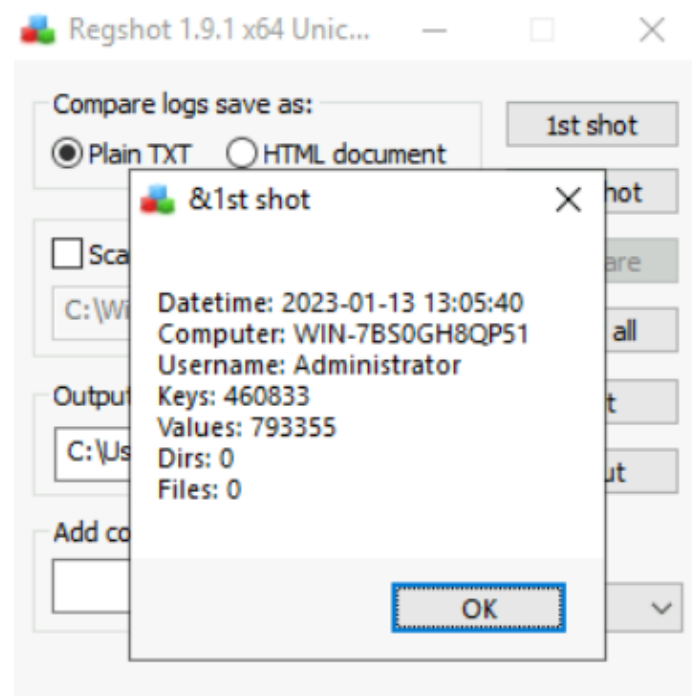
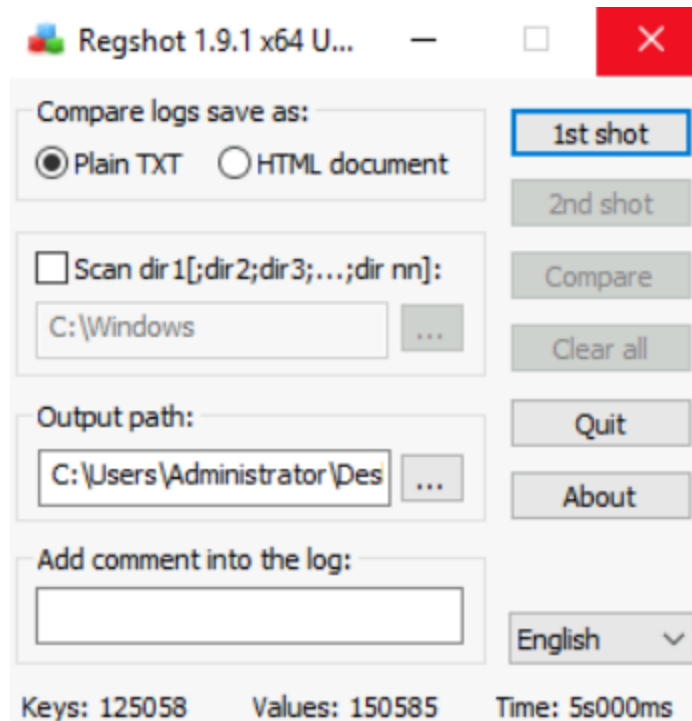
Is the file signed by a known organization? No

Is the process in the memory the same as the process on disk? No

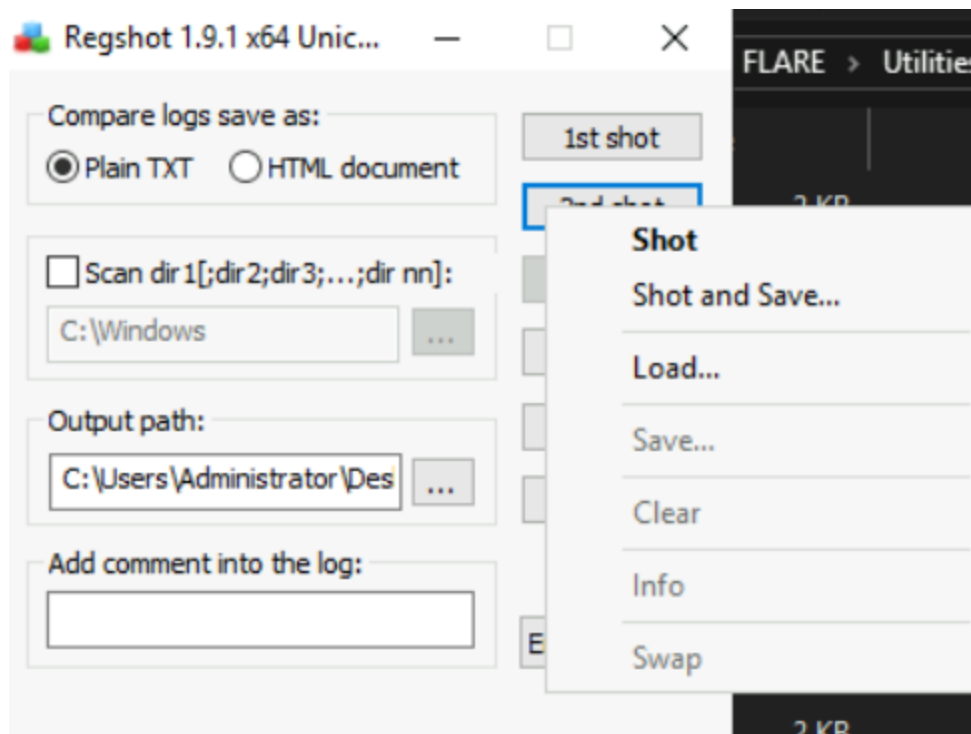
Task 6 - Regshot

Regshot is a tool that identifies any changes to the registry (or the file system we select). It can be used to identify what registry keys were created, deleted, or modified during our dynamic analysis by malware. Regshot works by taking snapshots of the registry before and after the execution of malware and then comparing the two snapshots to identify the differences between the two.

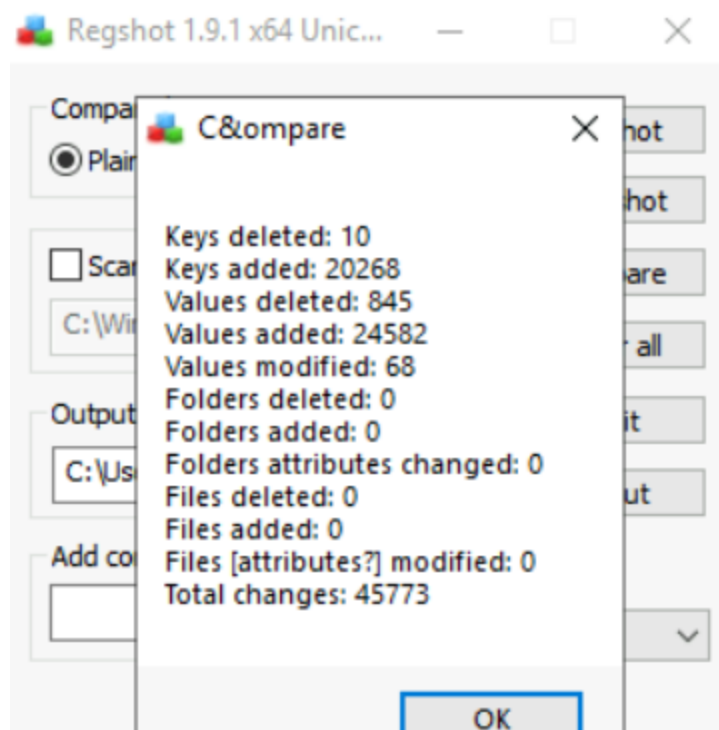
When we execute Regshot, we see the following interface. Please note that the Output path we see in the attached VM might differ.



Now that we have saved a shot of the registry, we can execute the malware. Once we have executed the malware and are confident that it has performed its malicious activity, we take a 2nd shot. For this, we click the '2nd shot' option.



Now that we have both shots, we can compare them to identify the registry changes performed by the malware. We do that by clicking the 'Compare' option. We will see a summary that looks something like the below screenshot.



Notice that it shows Keys and Values that were added, deleted, and modified. It also shows changes to Files and Folders. We see zero changes to Folders and Files because we had disabled 'Scan dir1' while taking the shots.

One advantage that Regshot enjoys over all the other tools discussed in this room is that it does not need to be running when we execute the malware. Some malware can check all the running processes and shut down if any analysis tool is running. When analyzing, we might often encounter malware samples that check for ProcExp, ProcMon, or API Monitor before performing any malicious activity and quitting if these processes are found. Therefore, these samples might thwart our analysis efforts. However, since Regshot takes a shot before and after the execution of the malware sample, it does not need to be running during malware execution, making it immune to this technique of detection evasion. On the flip side, we must ensure that no other process is running in the background while performing analysis with Regshot, as there is no filtering mechanism in Regshot, as we saw in the other tools. Hence, any noise created by background processes will also be recorded by Regshot, resulting in False Positives.

Task 7 - Conclusion

- Monitor a process's activities using ProcMon and filter out other processes to focus on the process of our interest.
- Identify what API calls a process is making to identify the behaviour of that process.
- Figure out if a malware sample is trying to evade detection by performing Process Masquerading or Process Hollowing. Process Explorer was our go-to tool for this purpose.
- Identify changes made by malware in the registry using Regshot.

Furthermore, we must understand that malware analysis requires perseverance, persistence, and attention to detail. Malware authors will always try to thwart an analyst's efforts, and what we have covered so far is not enough to analyze the most advanced malware.